

The Sociolinguistic Impact of Pretraining Data Curation: A Critical Analysis of the AboutMe Dataset and Filtering Heuristics

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1 Introduction

Large Language Models derive their capabilities—and their biases—from the composition of their pretraining data. Yet the curation pipelines that transform raw web crawls into training corpora remain opaque. Curators routinely deploy automated filters to remove “low-quality” text, enforce language constraints, and suppress toxic content, but these seemingly objective interventions impose strict linguistic norms that are rarely scrutinised (Lucy et al., 2024).

The ACL 2024 paper *AboutMe: Using Self-Descriptions in Webpages to Document the Effects of English Pretraining Data Filters* by Li Lucy et al. provides an empirical framework for quantifying these hidden mechanics. By grounding web text in the self-reported social and geographic contexts of its creators, the study demonstrates that data filters are not neutral arbiters of quality—they actively shape who is represented in foundation models (Lucy et al., 2024).

This report analyses the AboutMe methodology, the filter mechanisms it evaluates, and the sociolinguistic implications of current pretraining practices.

2 The AboutMe Dataset

2.1 Motivation

Sociolinguistic analysis of web corpora is hampered by a fundamental gap: raw crawl data lacks authorship metadata. Researchers cannot easily determine whether a demographic group’s language is being systematically excluded without first knowing who wrote what. The AboutMe dataset addresses this by using self-descriptive “About” pages as demographic anchors (Lucy et al., 2024).

2.2 Corpus Construction

The dataset draws on 24 public Common Crawl snapshots collected between May 2020 and June 2023, processed through the CCNet extraction pipeline (Wenzek et al., 2020). The construction follows three key steps:

Language filtering. Documents scoring above 0.5 on the fastText English language identification classifier were retained.

URL pattern matching. URLs containing paths such as `about`, `about-me`, `about-us`, or `bio` were identified as self-description pages. Only pages with a single candidate URL were kept, preferring HTTPS over HTTP.

Content pairing. Each “About” page was paired with a randomly sampled page from the same hostname. This dual structure allows demographic metadata from the biography to be used for evaluating filter behaviour on the creator’s general content.

The resulting corpus spans 10.3 million hostnames, yielding 3.1 billion tokens from self-descriptions and 3.5 billion tokens from paired content pages.

3 Extracting Creator Demographics

3.1 Topical Clustering

To map the topical landscape, “About” page text was vectorised using TF-IDF and clustered via balanced k -means with $k = 50$. Standard k -means tends to produce a few dominant clusters in skewed data—a phenomenon called *minority collapse*. Balanced k -means adds a size-equalisation constraint, dynamically adjusting penalties during iteration to ensure roughly equal cluster sizes (Malinen & Fränti, 2014). Although computationally more expensive ($O(n^2)$ in the assignment phase), this guarantees that the 50 surfaced topics—from software engineering to religious discourse—are equitably distributed, enabling robust downstream regression analyses.

3.2 Individual vs. Organisation Classification

Because corporate communications and personal blogs differ substantially in register, the methodology distinguishes individuals from organisations. Pages with explicitly singular URL paths (about-me) were labelled as individuals; collective paths (about-us) as organisations. For the ambiguous about path, a random forest classifier was trained on morphosyntactic features: pronoun ratios (singular vs. plural), NER tag frequencies (particularly PERSON entities from spaCy), and first-name occurrence rates. This classifier achieved a macro F1 of 89.2 under 5-fold cross-validation.

3.3 Occupation Extraction and Mapping

Occupation extraction was framed as sentence-level token classification, targeting self-identification expressions (e.g., “I am a freelance designer”). A RoBERTa-base model was first adapted via continued masked language modelling on the “About” page domain, then fine-tuned on 1,000 annotated sentences (Cohen’s $\kappa = 0.83$, word-level $F1 = 0.898$). The extracted roles were mapped to the O*NET-SOC taxonomy and cross-referenced with salary estimates and occupational prestige ratings from Hughes et al. (2022).

3.4 Geographic Inference

Geographic affiliations were inferred using Mordecai3, a neural geoparsing library (Halterman, 2023). The pipeline works in three stages: (1) toponym recognition via spaCy’s transformer NER model; (2) candidate retrieval from a GeoNames gazetteer hosted in Elasticsearch; and (3) neural ranking using RoBERTa-based similarity embeddings, with an abstention capability for ambiguous cases. Mordecai3 achieves 79–94% exact match accuracy depending on the evaluation corpus, and successfully tagged 63.2% of hostnames with country-level designations.

4 The Filters Under Evaluation

The study evaluates ten filters across four categories, summarised in Table 1.

4.1 Model-Based Quality Classifiers

These filters train binary logistic regression models where the positive class is a curated reference corpus and the negative class is random Common Crawl text. Documents scoring below a threshold are discarded. The choice of reference corpus determines what counts as “quality”: WIKI favours encyclopaedic prose, OPENWEB

Table 1: Pretraining data filters evaluated in the AboutMe study.

Category	Filter	Mechanism	Reference Data
4*Quality	WIKI	Logistic regression	Wikipedia
	WIKIWEBBOOKS	Logistic regression	Wikipedia + Books3
	OPENWEB	Logistic regression	OpenWebText (Reddit)
	WIKIREFS	Logistic regression	Wikipedia references
Perplexity	WIKI _{PPL}	5-gram KenLM	Wikipedia
Heuristic	Gopher	19 deterministic rules	N/A
4*LangID	fastText	Char n -gram + hier. softmax	Wikipedia, Tatoeba
	CLD2	Naive Bayes	Web text
	CLD3	Neural n -gram hashing	Multilingual web
	langdetect	Probabilistic profiles	Diverse text

favours Reddit-popular content, and WIKIREFS favours academic citations.

4.2 Perplexity-Based Filtering

The WIKI_{PPL} filter, based on the CCNet pipeline (Wenzek et al., 2020), trains a 5-gram Kneser-Ney language model on Wikipedia using KenLM. Documents are tokenised with SentencePiece, scored by perplexity, and segmented into terciles. Low-perplexity text (structurally similar to Wikipedia) is retained; high-perplexity text (colloquial, fragmented, or domain-specific) is pruned. Wenzek et al. showed that BERT models trained on the low-perplexity partition outperformed Wikipedia-trained equivalents by 3.3% on XNLI benchmarks.

4.3 Heuristic Filters

The Gopher heuristics (Rae et al., 2021) apply 19 deterministic rules targeting formatting errors, repetition, and non-natural language. These include constraints on document length (50–100,000 words), mean word length (3–10 characters), alphabetical character proportions, duplicate line fractions, and minimum stop-word frequencies. Documents failing any rule are removed entirely.

4.4 Language Identification

Four LangID systems were evaluated. fastText uses character n -grams with hierarchical softmax to identify 170+ languages. CLD2 is a Naive Bayes classifier optimised for web/HTML text. CLD3 hashes character n -grams into dense embeddings processed by a multi-layer perceptron. langdetect is a Python port of Google’s legacy Java library covering 55 languages. All operate under a monolingual assumption—they presume

each document exists in a single language.

5 Key Findings

5.1 Quality Filters as Occupational Gatekeepers

The most striking finding is that quality classifiers behave as implicit topical filters. Because they learn to prefer text resembling their reference corpora, they systematically penalise topics and occupations absent from those domains.

OPENWEB shows massive preference for technology and programming occupations—unsurprising given its Reddit provenance. WIKIREFS favours journalism and academia. Across nearly all quality classifiers, commercial and lifestyle content (online shopping, fashion, manual trades) suffers the highest removal rates.

OLS regression with standardised z -scores revealed a statistically significant relationship ($p < 0.05$) between occupational prestige and filter retention. Roles such as “barista,” “home inspector,” and “maid” were removed at substantially higher rates than high-prestige occupations. This bias is an artefact of the reference data: encyclopaedias and curated book corpora naturally skew toward educated, high-status populations, causing the classifier to treat working-class registers as low quality.

5.2 Lexical Amplification

The study identified a compounding effect: the degree to which a page is retained or removed is amplified by role-specific terminology. If a filter already favours an occupation, dense use of that occupation’s vocabulary further boosts retention scores. Conversely, if a filter penalises an occupation, its specialised jargon compounds the penalty. The filters are not merely targeting poor syntax—they are suppressing the professional lexicons of blue-collar and commercial domains.

5.3 Geographic Exclusion Through LangID

LangID systems introduce severe geographic disparities through their monolingual assumption. On the multilingual global internet, writers routinely embed local names, geographic entities, and culturally specific terms within English text. LangID n -gram models interpret these non-Western tokens as foreign vocabulary, artificially lowering the document’s English probability score.

The disparities are substantial: content from Eastern Asian creators was 2.4 times more likely to be removed by CLD2 than content from Northern Europe. Even after control-

ling for topical domain in OLS regressions, the Asian subregion exhibited the most extreme negative coefficients across all four LangID systems. The net effect is an artificial “Core Anglophone” hegemony in training data.

6 The Toxicity Filtering Dilemma

These findings must be contextualised alongside toxicity filtering. Longpre et al. (2023) trained 28 decoder-only models (1.5B parameters each) on varying data permutations and established a direct trade-off: aggressive toxicity filtering lowers the risk of toxic outputs but simultaneously degrades downstream generalisation performance (Longpre et al., 2023).

Most critically, models trained on heavily sanitised data perform *worse* at identifying toxicity in downstream tasks—they lack the representational capacity to understand adversarial prompts or contextual harm. Longpre et al. found that “inverse toxicity filtering” (retaining more toxic text) was paradoxically the most effective strategy for building toxicity *detection* capability.

Toxicity classifiers like Google’s Perspective API, which uses distilled BERT-based CNNs trained on crowdsourced annotations, also exhibit well-documented false-positive rates for marginalised identities. African American English and reclaimed slurs within minority communities are routinely misclassified as offensive. Combined with the quality and LangID biases documented by AboutMe, the cumulative effect is a multi-layered erasure of working-class, regional, and marginalised discourse from training data.

7 Broader Implications

7.1 Epistemic Homogenisation

The widespread use of Wikipedia as reference data for quality classifiers, perplexity models, and even LangID training instils a specific epistemic bias. Wikipedia’s editorial norms—notability, neutral point of view, academic sourcing—become the implicit standard for “quality.” This produces models that default to encyclopaedic prose while struggling with conversational, commercial, or colloquial registers.

7.2 The Alignment Paradox

Post-training alignment techniques such as RLHF require the base model to possess internal representations of the concepts being aligned. If LangID filters erase code-

switching, quality filters erase working-class lexicons, and toxicity filters erase minority dialects, the model cannot be effectively aligned for those populations. The pruned concepts are entirely out-of-distribution, making the model brittle and prone to hallucination for diverse users.

7.3 Limitations of Rule-Based Heuristics

Even the Gopher heuristics, designed to bypass ML classifier biases through objective thresholds, disproportionately affect certain domains. Short, contextual descriptions from visual artists may fail length or stop-word requirements, while verbose SEO-optimised spam passes easily. At trillion-token scale, blunt mathematical rules cannot capture the structural diversity of human language.

8 Towards Better Curation Practices

The AboutMe findings point toward three strategic shifts:

Routing instead of pruning. Rather than discarding high-perplexity or low-scoring data, pipelines can use balanced clustering to segment crawls into topical domains and train models via domain-specific routing (e.g., Mixture-of-Experts or Branch-Train-Merge), ensuring diverse domains are modelled rather than deleted.

Context-aware language identification. LangID systems should employ sliding-window evaluation or entity masking—suppressing named entities before scoring—to prevent valid global English from being flagged as non-English due to embedded local terminology.

Demographic metadata tracking. The AboutMe methodology (NER, geoparsing, occupation extraction) can be integrated into curation pipelines as a live metadata ledger, enabling active demographic balancing before training begins.

9 Conclusion

The transformation of web crawls into training corpora is a deeply subjective process masked by algorithmic objectivity. The AboutMe study demonstrates that quality classifiers function as gatekeepers of occupational prestige, perplexity filters enforce encyclopaedic homogenisation, and LangID systems marginalise global English variants. Combined with aggressive toxicity filtering, the result is training data reflecting a heavily skewed WEIRD (Western, Educated, Industrialised, Rich, Democratic) perspective. Building LLMs that serve a global constituency requires evolving data cura-

tion into a sociolinguistically aware discipline—one that replaces blunt pruning with representation-conscious engineering.

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